
hrHSA

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hrHSA (*high-resolution Habitat Selection Analysis*) is a scalable Python framework for rigorous statistical inference on spatially distributed observations. It integrates positional data with environmental covariates to quantify how heterogeneous spatial fields, individual variation, and changing conditions shape the probability distribution of observed locations. The framework combines resource-selection analysis, spatial point-process models, hierarchical Bayesian inference, and mechanistic simulation, with explicit treatment of availability, observation processes, uncertainty, validation, and prediction.

Developed primarily for wildlife telemetry and spatial ecology, **hrHSA** addresses the more general problem of inferring interactions between actively and/or passively redistributing entities and their spatial context. Its methods therefore extend naturally to applications in geospatial science, epidemiology, environmental modelling, human mobility, and other domains concerned with spatially structured processes. Reproducible, vectorized, out-of-core, and HPC-ready workflows support analyses from covariate extraction and parameter estimation to predictive simulation.

GETTING STARTED

This guide introduces the core **frequentist habitat-selection workflow** in hrHSA. Starting from telemetry records and a raster stack of environmental covariates, we will:

1. prepare and project relocation data;
2. define the spatial domain considered available;
3. sample environmental conditions at used and available locations;
4. inspect predictor distributions;
5. compare a biologically defined set of candidate models;
6. fit and project a resource-selection function;
7. evaluate transferability among individuals.

The final section briefly shows how the same used–available design can be extended to hierarchical Bayesian inference.

1.1 Installation

Clone the repository and install hrHSA with its geospatial and distributed-computing dependencies:

```
git clone <repository-url>
cd hrHSA
python -m pip install -e "[hpc]"
```

Import the components used throughout this guide:

```
import numpy as np
import pandas as pd

from hsa import FeatureSpec
from hsa.movement.geometry import prepare_trajectory_data
from hsa.compute import open_raster_stack_zarr, suggest_xy_chunks
from hsa.sampling import (
    get_availability_domain,
    sample_available_points,
    sample_raster_stack,
)
from hsa.diagnostics import inspect_sampled_use_available
from hsa.rsfc import (
    compare_single_predictors,
    evaluate_linear_candidates_up_to_k,
```

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```

    fit_rsf,
    predict_rsf_surface,
)
from hsa.rsf.cv import leave_one_individual_out_rsf
from hsa.rsf.validation import plot_boyce_curves

```

1.2 1. Prepare relocation data

hrHSA expects timestamps, individual identifiers, and spatial coordinates. Relocations should be represented in a projected coordinate reference system appropriate for the study area.

```

raw = pd.read_csv("data/relocations.csv")

reloc = prepare_trajectory_data(
    raw,
    id_col="Individual_ID",
    timestamp_col="Timestamp",
    lon_col="longitude",
    lat_col="latitude",
    source_crs="EPSG:4326",
    target_crs="EPSG:32733", # replace with the CRS of the raster stack
    round_freq="h",
    drop_duplicate_fixes=True,
)

reloc.head()

```

The function parses timestamps, removes invalid records, projects coordinates, sorts trajectories, and optionally removes duplicate observations within a specified temporal interval.

Important

The target CRS must use meaningful planar units and should match the environmental raster stack. Geographic longitude–latitude coordinates should not be used for distance-, area-, or availability-based operations.

1.3 2. Open the environmental raster stack

Environmental covariates can be stored as a Zarr-backed `xarray.DataArray` with dimensions `band`, `y`, and `x`.

```

env = open_raster_stack_zarr(
    "data/environment.zarr",
    name="env",
    chunks=None,
)

chunks = suggest_xy_chunks(
    env,
    target_chunk_mb=128,
)

```

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```
env = env.chunk(chunks)

print(chunks)
print(env)
```

`suggest_xy_chunks` chooses approximately square spatial chunks while retaining all predictor bands within each chunk. This provides a defensible starting point for raster sampling and prediction, although chunk sizes should ultimately be benchmarked against the available memory, storage system, and workload.

Confirm that the relocation and raster coordinate systems agree:

```
assert reloc.crs == env.rio.crs
```

1.4 3. Define availability

A resource-selection analysis compares observed use with environmental conditions that were accessible but not necessarily used. The definition of availability is therefore part of the ecological model rather than merely a computational preprocessing decision.

Here, separate availability domains are estimated for each individual using a centroid-trimmed 95% minimum convex polygon:

```
domains = get_availability_domain(
    reloc,
    id_col="Individual_ID",
    estimator="mcp",
    quantile=0.95,
    min_points=5,
)

domains.head()
```

Available locations are then sampled independently within each individual domain:

```
samples = sample_available_points(
    domains,
    used=reloc,
    id_col="Individual_ID",
    n_per_used=10,
    seed=42,
)

samples["used"].value_counts()
```

The resulting table contains both observed relocations (`used=True`) and sampled available locations (`used=False`).

Note

The number of available points controls the numerical approximation of the available environment. It does not represent the number of locations an animal could literally have visited.

1.5 4. Extract environmental covariates

Sample all raster bands at the used and available locations:

```
sampled = sample_raster_stack(  
  samples,  
  env,  
  id_cols="Individual_ID",  
)  
  
sampled.head()
```

Each row now represents one used or available location together with its environmental covariates.

1.6 5. Inspect used and available environments

Before fitting a model, inspect missingness, predictor distributions, and the degree of separation between used and available samples.

```
inspection = inspect_sampled_use_available(  
  sampled,  
  id_col="Individual_ID",  
  plot=True,  
  plot_top_n=8,  
)  
  
inspection["pooled_stats"]
```

The diagnostic summary includes distributional differences such as the Kolmogorov–Smirnov statistic, Wasserstein distance, differences in means and medians, and the common-language effect size.

These summaries are descriptive. They are useful for identifying data problems, implausible contrasts, and potentially informative predictors, but they do not replace a defined ecological hypothesis or multivariable model.

1.7 6. Evaluate candidate models

Begin with a biologically justified set of candidate predictors:

```
candidate_predictors = [  
  "elevation",  
  "slope",  
  "ndvi",  
  "distance_to_water",  
)
```

Compare single-predictor models:

```
single_predictor_models = compare_single_predictors(  
  sampled,  
  predictors=candidate_predictors,  
)  
  
single_predictor_models[
```

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```
[
  "predictor",
  "AIC",
  "delta_aic",
  "akaike_weight",
  "pseudo_r2",
  "converged",
]
```

Small candidate combinations can then be evaluated:

```
candidate_models = evaluate_linear_candidates_up_to_k(
  sampled,
  predictor_cols=candidate_predictors,
  max_k=3,
  n_jobs=4,
)

candidate_models.head(10)
```

Warning

The number of possible models grows combinatorially. Candidate predictors, transformations, and interactions should be constrained by ecological reasoning before model fitting. AIC ranking quantifies relative support within the supplied candidate set; it does not discover causal structure automatically.

1.8 7. Specify and fit the RSF

FeatureSpec defines the model matrix explicitly. Continuous predictors are standardized during fitting, while quadratic terms and interactions are constructed on the standardized scale.

```
spec = FeatureSpec(
  linear=[
    "distance_to_water",
    "ndvi",
    "slope",
  ],
  quadratic=[
    "distance_to_water",
  ],
  interactions=[
    ("ndvi", "slope"),
  ],
  add_const=True,
)
```

Fit the used-available logistic regression:

```
model, scaler, fitted_spec, meta = fit_rsf(
    sampled,
    spec,
)

print(model.summary())
```

For continuous predictors, positive coefficients indicate increasing relative selection with increasing predictor values, conditional on the remaining terms. Negative coefficients indicate decreasing relative selection.

Because the model is fitted to a researcher-defined ratio of used and available locations, its intercept and raw logistic probabilities generally do not represent absolute occurrence probabilities. The primary inferential quantities are relative selection coefficients and contrasts in relative selection strength.

1.9 8. Predict the selection surface

Project the fitted model over the environmental raster stack:

```
rsf = predict_rsf_surface(
    env,
    model,
    scaler,
    fitted_spec,
    meta,
)

rsf
```

The output is an `xarray.DataArray` containing the relative selection surface:

```
rsf.sel(band="rsf").plot(
    robust=True,
    figsize=(9, 7),
)
```

The surface is proportional to

$$[w(\mathbf{x}) = \exp(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})]$$

and should therefore be interpreted comparatively: a location with a score of two is estimated to have twice the relative selection strength of a location with a score of one, under the fitted model and availability design.

1.10 9. Validate among individuals

Randomly splitting autocorrelated relocations can produce optimistic validation results. For multi-individual datasets, hrHSA therefore supports leave-one-individual-out validation.

```
(
    cv_summary,
    cv_parameters,
    boyce_bins,
    calibration_bins,
    cv_diagnostics,
```

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```

) = leave_one_individual_out_rsf(
    reloc,
    env,
    fitted_spec,
    id_col="Individual_ID",
    heldout="all",
    domain_quantile=0.95,
    thin_train_dt="12h",
    sampling_factor_train=10,
    n_background_boyce=100_000,
    seed=42,
)

cv_summary

```

In each fold, the model is trained on all but one individual and evaluated against the relocations and availability domain of the held-out individual. This tests whether population-level selection relationships transfer to animals not used during model fitting.

Plot the fold-specific Boyce curves:

```

fig, ax = plot_boyce_curves(
    boyce_bins,
    id_col="heldout_ID",
)

```

The Boyce diagnostic evaluates whether observed locations become progressively more frequent than expected under random use as predicted relative selection increases. It should be interpreted alongside fold-specific coefficients, calibration diagnostics, sample sizes, and the ecological structure of the held-out individuals.

1.11 A hierarchical Bayesian extension

The frequentist workflow estimates a common population-level selection relationship. When repeated observations from multiple individuals are available, a hierarchical model can estimate both population-level effects and among-individual variation.

The following abbreviated PyMC model gives each individual its own intercept and slope while partially pooling those effects toward the population mean.

```

import pymc as pm

predictor = "distance_to_water"

bayes_df = sampled[
    ["used", "Individual_ID", predictor]
].dropna().copy()

x = bayes_df[predictor].to_numpy(dtype=float)
x = (x - x.mean()) / x.std()

y = bayes_df["used"].to_numpy(dtype=int)

id_idx, individual_names = pd.factorize(

```

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```

    bayes_df["Individual_ID"],
    sort=True,
)

coords = {
    "obs": np.arange(len(bayes_df)),
    "individual": individual_names.tolist(),
}

```

```

with pm.Model(coords=coords) as hierarchical_rsf:

    # Data containers
    x_data = pm.Data("x", x, dims="obs")
    id_data = pm.Data("id_idx", id_idx, dims="obs")
    y_data = pm.Data("y", y, dims="obs")

    # Population-level intercept and slope
    alpha = pm.Normal("alpha", mu=0.0, sigma=2.5)
    beta = pm.Normal("beta", mu=0.0, sigma=1.0)

    # Among-individual variation
    sigma_alpha = pm.Exponential("sigma_alpha", lam=1.0)
    sigma_beta = pm.Exponential("sigma_beta", lam=1.0)

    # Standardized individual deviations
    z_alpha = pm.Normal(
        "z_alpha",
        mu=0.0,
        sigma=1.0,
        dims="individual",
    )
    z_beta = pm.Normal(
        "z_beta",
        mu=0.0,
        sigma=1.0,
        dims="individual",
    )

    # Non-centered random effects
    alpha_ind = pm.Deterministic(
        "alpha_ind",
        z_alpha * sigma_alpha,
        dims="individual",
    )
    beta_ind = pm.Deterministic(
        "beta_ind",
        z_beta * sigma_beta,
        dims="individual",
    )

    # Total individual effects
    alpha_total = pm.Deterministic(

```

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```

    "alpha_total",
    alpha + alpha_ind,
    dims="individual",
  )
  beta_total = pm.Deterministic(
    "beta_total",
    beta + beta_ind,
    dims="individual",
  )

  # Linear predictor
  eta = (
    alpha_total[id_data]
    + beta_total[id_data] * x_data
  )

  # Used-available likelihood
  used = pm.Bernoulli(
    "used",
    logit_p=eta,
    observed=y_data,
    dims="obs",
  )

  inference = pm.sample(
    draws=2_000,
    tune=2_000,
    chains=4,
    target_accept=0.95,
    random_seed=42,
  )

```

The population slope `beta` describes the average selection response, while `sigma_beta` quantifies the degree to which that response varies among individuals. Partial pooling regularizes poorly sampled individuals and propagates uncertainty through population- and individual-level estimates.

As in the frequentist used–available model, the intercept depends on the availability-sampling design. Individual slopes and their population distribution are generally the principal ecological targets.

Note

The hierarchical model is currently presented as an optional extension rather than a stabilized hrHSA interface. A future Bayesian module will connect design-matrix construction, prior specification, posterior diagnostics, spatial prediction, and posterior predictive validation within the package workflow.

THEORY

2.1 1. Foundations

2.1.1 Habitat

Habitat provides the ecological link between individual movement and population redistribution. By remaining, departing, returning or continuing onwards, animals allocate their time among heterogeneous environments. Repeated over time, these decisions produce patterns of local space use, home-range formation, migration and seasonal range occupation; aggregated across individuals, they generate population-level patterns of distribution and density. Habitat selection therefore connects the relocation of the individual with the organisation of the population (Fretwell & Lucas, 1969; Morris, 2003; Nathan et al., 2008; Mueller & Fagan, 2008; Matthiopoulos et al., 2015).

Within Habitat Selection Analysis, *habitat* is most usefully understood as a point in environmental space defined by the conjunction of conditions, resources and risks relevant to the focal organism (Morris, 2003; Matthiopoulos et al., 2020; Northrup et al., 2022). Conditions influence physiological or behavioural functioning; resources are required for maintenance, growth or reproduction; and risks reduce expected survival or reproductive success. Habitat is therefore organism-specific: the same geographic location may present different opportunities and constraints to individuals differing in state, experience or perceptual capacity.

Several related terms must be distinguished. *Use* is the realised allocation of an animal's time or activity to a habitat. *Occupancy* denotes physical presence within a spatial unit during a defined period. *Availability* describes the habitats that could have been encountered or used, given the spatial, temporal and behavioural constraints acting upon the animal. *Selection* is differential use relative to that availability, whereas *preference*, in its strict sense, describes selection where all alternatives are equally available (Johnson, 1980; Manly et al., 2002; Lele et al., 2013).

For habitat types $h = 1, \dots, H$, let a_h denote the proportion of available habitat belonging to type h , and let u_h denote its proportion of observed use. For $a_h > 0$, the elementary selection ratio is

$$r_h = \frac{u_h}{a_h}.$$

Use is proportional to availability when $r_h = 1$, greater than expected when $r_h > 1$, and less than expected when $r_h < 1$. Selection is thus a relationship between use and availability, not an intrinsic property of the habitat. A widespread habitat may contain many recorded locations while receiving less use than expected from its abundance; a rare habitat may contain few observations while nevertheless receiving strongly disproportionate use.

2.1.2 The Ideal Free Distribution

Although selection is enacted by individuals, habitat value depends partly upon the population already occupying it. Competitors may deplete resources, interfere with foraging, defend territories or exclude subordinate animals. Population distribution is therefore both a consequence of earlier habitat-selection decisions and part of the ecological context governing subsequent ones. Let N_h denote the number or density of animals occupying habitat h , and let $\Phi_h(N_h)$ denote the expected per-capita fitness payoff available there. Under competition,

$$\frac{\partial \Phi_h(N_h)}{\partial N_h} < 0,$$

so that habitat value declines as density increases. Habitat selection is therefore inherently density dependent (Fretwell & Lucas, 1969; Rosenzweig, 1981; Morris, 2003; McLoughlin et al., 2010).

The classical formulation of this relationship is the **Ideal Free Distribution** of Fretwell and Lucas (1969). The term *ideal* assumes that individuals can distinguish the relative payoffs of available habitats; the term *free* assumes that they may move among them without prohibitive costs or exclusion. Consider a population of total size N distributed among H habitats, with N_h^* individuals occupying habitat h at equilibrium:

$$\sum_{h=1}^H N_h^* = N.$$

An ideal-free equilibrium is reached when no individual could improve its expected fitness by moving elsewhere. All occupied habitats therefore provide the same equilibrium payoff λ :

$$\Phi_h(N_h^*) = \lambda \quad \text{for all } h \text{ with } N_h^* > 0,$$

while any unoccupied habitat offers no greater return:

$$\Phi_h(0) \leq \lambda \quad \text{when } N_h^* = 0.$$

A habitat with a high initial payoff may consequently be occupied first, but the arrival of additional individuals reduces its per-capita value. Once its payoff falls to that available elsewhere, individuals distribute themselves among habitats in the proportions required to maintain equal expected fitness. Habitats of greater productive capacity may therefore support greater animal densities without providing greater realised fitness at equilibrium: their greater density is precisely what reduces the individual payoff to that available in alternative habitats.

For two habitats, combinations of densities satisfying

$$\Phi_1(N_1) = \Phi_2(N_2)$$

form an **isodar** (Morris, 1987, 1988, 2003). The isodar intercept represents differences in habitat profitability at low density, whereas its slope represents differences in the rate at which fitness declines with crowding. Ideal-free and isodar theory thus provide a formal bridge between individual habitat choice, density-dependent competition and population distribution.

2.1.3 The Ideal Despotic Distribution

The assumptions of the Ideal Free Distribution provide a theoretical reference condition rather than a universal description of nature. Individuals differ in information, state and competitive ability; movement may be costly; and dominant animals may monopolise profitable habitats. Such processes may produce an **Ideal Despotic Distribution**, in which subordinate individuals occupy less profitable habitats despite the existence of superior alternatives (Fretwell & Lucas, 1969). Observed distributions may therefore reflect habitat profitability, competition, restricted access and social organisation in combination.

Habitat selection is also hierarchical and scale dependent. Johnson (1980) distinguished selection of a species' geographical range, placement of an individual home range within that range, use of habitat components within the home range and selection of particular resources within an activity site. A habitat may be selected at one order and avoided at another, or selected for one activity and avoided during another. Availability is therefore an ecological hypothesis concerning the alternatives accessible to an animal at a particular time and scale, rather than a purely geometric property of the study area.

Habitat Selection Analysis begins from these premises. Animals respond to environmental heterogeneity through movement and space use; their decisions produce population distributions; and the resulting densities alter the value of habitats subsequently encountered. The inferential question that follows is how this relationship may be recovered from observations of animal relocations and measured environmental features.

2.2 2. Making it real: Towards Inference from Empirical Data

2.2.1 From Relocations to Selection

Habitat selection is a behavioural process, but telemetry records only a finite sample of its spatial consequences. Let $S_i(t)$ denote the location of individual i at time t . A tracking device observes this continuous trajectory only at discrete times t_{ij} , producing the relocation series

$$\mathcal{D}_i = \{\tilde{s}_{ij}, t_{ij}\}_{j=1}^{n_i},$$

where $\tilde{s}_{ij}$ may differ from the true location because of positional error. The observed sample is further conditioned upon the fix schedule and the probability of successfully recording a location.

A concentration of relocations in one part of the landscape may therefore arise because the habitat is selected, because it is readily accessible, because the animal moves slowly within it, or because it repeatedly returns to the same area. Conversely, few relocations may indicate avoidance, restricted access, rapid passage or reduced fix success. Selection cannot be inferred from observations of use alone; the observed distribution must be compared with an explicit model of what was available to the animal.

2.2.2 Resource Selection as a Point Process

Consider a spatial domain $\Omega \subset \mathbb{R}^2$ containing observed relocations

$$\mathcal{S} = \{s_1, \dots, s_n\}.$$

Spatial use may be represented by an **inhomogeneous Poisson point process**. If $N(B)$ denotes the number of relocations observed within a region $B \subseteq \Omega$, then

$$N(B) \sim \text{Poisson} \left(\int_B \lambda(s) ds \right),$$

where $\lambda(s) > 0$ is the spatial intensity of relocations.

Environmental heterogeneity is commonly introduced through a log-linear intensity:

$$\lambda(s) = \exp[\beta_0 + x(s)\beta],$$

where $x(s)$ is a row vector of environmental covariates and β is the corresponding vector of coefficients. This may be written as

$$\lambda(s) = \exp(\beta_0) w(x(s)),$$

with the selection function

$$w(x) = \exp(x\beta).$$

The function $w(x)$ describes how habitat alters the relative intensity of use. It is not the absolute probability that a spatial unit will be occupied or used.

Where locations differ in their accessibility, the intensity may be written more generally as

$$\lambda(s) = c f^A(s) w(x(s)),$$

where $f^A(s)$ is the available distribution and $c > 0$ controls the expected number of observations. Conditional upon observing n relocations, the corresponding use distribution is

$$f^u(s) = \frac{f^A(s) w(x(s))}{\int_{\Omega} f^A(r) w(x(r)) dr}.$$

The available environment is thus reweighted by the selection function to produce the expected distribution of use.

The intercept has little direct biological meaning in a conventional telemetry analysis because the total number of relocations depends upon the fix schedule, study duration and number of tracked animals. The coefficients in β , by contrast, describe relative differences in the intensity of use, conditional upon the defined available distribution.

2.2.3 Approximation by Logistic Regression

The likelihood of an inhomogeneous Poisson point process contains an integral over the full spatial domain:

$$L(\beta) \propto \exp \left[- \int_{\Omega} \lambda(s) ds \right] \prod_{i=1}^n \lambda(s_i).$$

For realistic landscapes, this integral must usually be approximated numerically. In a use–availability design, environmental conditions are evaluated at the observed relocations and at locations sampled from the available distribution.

Let

$$Y_j = \begin{cases} 1, & \text{if location } j \text{ is used,} \\ 0, & \text{if location } j \text{ is sampled from availability.} \end{cases}$$

The combined sample may be fitted using logistic regression:

$$\text{logit} [\Pr(Y_j = 1 \mid x_j)] = \alpha + x_j \beta.$$

Available locations are not biological absences. They represent the environmental distribution from which the animal could have selected and provide a numerical approximation to the point-process integral. As the available sample becomes sufficiently large, the logistic slopes approach those of the underlying point-process model (Johnson et al., 2006; Warton & Shepherd, 2010; Aarts et al., 2012).

The logistic intercept α depends upon the analyst-defined ratio of used to available locations and should not ordinarily be interpreted. Predictions should therefore be based upon the exponential selection function

$$\hat{w}(x) = \exp(x\hat{\beta}),$$

rather than the fitted logistic probability

$$\frac{\exp(\alpha + x\hat{\beta})}{1 + \exp(\alpha + x\hat{\beta})}.$$

The number of available locations should be increased until the estimated slopes stabilise. Additional available points improve numerical integration but do not create additional biological replication.

2.2.4 Interpretation of an RSF

For two environmental conditions x_1 and x_0 , their relative selection strength is

$$\text{RSS}(x_1, x_0) = \frac{\hat{w}(x_1)}{\hat{w}(x_0)} = \exp \left[(x_1 - x_0)\hat{\beta} \right].$$

For two otherwise identical locations differing by one unit in covariate x_j ,

$$\text{RSS} = \exp(\hat{\beta}_j).$$

Values greater than one indicate greater relative selection for the first condition; values below one indicate lower relative selection. These contrasts remain conditional upon the other model terms, the available domain and the scale of analysis.

Interactions, transformations and nonlinear effects should be interpreted through explicit contrasts rather than through isolated coefficients. A mapped RSF surface,

$$\hat{w}(x(s)) = \exp \left[x(s)\hat{\beta} \right],$$

has an arbitrary absolute scale. It may be normalised within a defined available domain to obtain

$$\hat{f}^u(s) = \frac{f^A(s)\hat{w}(x(s))}{\int_{\Omega} f^A(r)\hat{w}(x(r)) dr},$$

but the resulting distribution is specific to that domain.

2.2.5 Individuals and Autocorrelation

Telemetry studies may contain many relocations but comparatively few animals. Locations from the same individual are repeated observations of one behavioural process and should not be treated as independent population replicates.

Individual variation may be represented hierarchically:

$$w_i(x) = \exp(x\beta_i),$$

with

$$\beta_i \sim \mathcal{N}(\mu, \Sigma),$$

where μ describes the population-level response and Σ describes variation among individuals. Random slopes are important because animals may differ in the direction or strength of selection. A random intercept alone does not represent such behavioural variation.

Successive relocations are also serially dependent. An animal's current position constrains its next position, and nearby observations commonly share similar environmental conditions. Ignoring this dependence generally leads to underestimated uncertainty.

Autocorrelation should not automatically be removed by thinning the data until successive points appear independent. Residence, return and slow movement may themselves be biologically meaningful. More defensible approaches include hierarchical models, robust standard errors, block bootstrapping, two-stage analyses and structured cross-validation.

2.2.6 Model Evaluation

Evaluation should be based upon held-out data and should reflect the intended form of prediction. Interpolation within known individuals, prediction to later periods and transfer to new individuals are distinct tasks.

A commonly used measure for presence-only or use-availability predictions is the **continuous Boyce index**. Let $\hat{w}(s)$ denote the predicted RSF value in an evaluation dataset. The prediction range is divided into intervals or moving windows I_k . For each interval, calculate the proportion of held-out used locations,

$$P_k = \frac{\text{held-out used locations in } I_k}{\text{all held-out used locations}},$$

and the proportion of available locations,

$$E_k = \frac{\text{available locations in } I_k}{\text{all available locations}}.$$

Their ratio is

$$R_k = \frac{P_k}{E_k}.$$

The Boyce index is the Spearman rank correlation between the representative prediction value v_k and the predicted-to-expected ratio:

$$B = \text{cor}_S(v_k, R_k).$$

Values approaching 1 indicate that greater predictions correspond to greater use relative to availability. Values near 0 indicate little monotonic relationship, while negative values indicate that held-out locations occur disproportionately in areas assigned low predictions.

The Boyce index evaluates ranking rather than complete probabilistic calibration. It does not preserve the magnitude or geographic location of prediction errors and may depend upon window width, ties and sample size. It should therefore be accompanied by the full predicted-to-expected curve, observed and expected use across prediction quantiles, and spatial inspection of held-out residual patterns.

2.2.7 Structured Cross-Validation

Randomly dividing relocations among folds is generally unsuitable for telemetry data. Neighbouring observations from the same movement bout may enter both the training and evaluation sets, producing information leakage and overly optimistic performance estimates.

For prediction to another period within the same animals, data should be divided into contiguous **temporal blocks**. Blocks should be sufficiently long to reduce dependence between training and evaluation observations. Where forecasting is the objective, models should be fitted to earlier periods and evaluated on later ones.

For inference to the wider population, **leave-one-individual-out cross-validation** is more appropriate. All relocations from one animal are withheld, the model is fitted to the remaining animals, and its predictions are evaluated against the held-out individual. Repeating this procedure assesses whether the estimated population response transfers to animals not represented during fitting.

Where both temporal transfer and population transfer matter, the two schemes may be combined. An outer leave-one-individual-out loop can assess generalisation to new animals, while temporally blocked validation within the training animals can guide model development.

Validation scores should be reported by individual and fold. A single pooled score may be dominated by animals contributing the greatest number of relocations.

2.2.8 From Resource Selection to Step Selection

A conventional RSF usually compares every relocation with a common availability distribution, such as an individual's home range or the study area. At fine temporal scales, this assumption becomes difficult to defend. A location may lie within the home range but remain unreachable during the interval between two fixes.

Movement therefore determines availability. The locations accessible at time $t + 1$ depend upon the animal's position at time t , the elapsed time, its movement capacity and its previous direction of travel.

A **step** is the displacement between consecutive relocations:

$$s_t \longrightarrow s_{t+1}.$$

Its length is

$$l_t = \|s_{t+1} - s_t\|,$$

and its turning angle θ_t is the change in bearing relative to the preceding step.

In a **Step Selection Function**, each observed step is paired with available steps beginning at the same location. Their lengths and turning angles are conventionally sampled from the empirical distributions of observed movements. Habitat conditions at the observed endpoint or along the observed path are then compared with those of the available alternatives.

Let

$$\mathcal{C}_t = \{s_{t+1}^{(0)}, s_{t+1}^{(1)}, \dots, s_{t+1}^{(K)}\}$$

denote the choice set for step t , where $s_{t+1}^{(0)}$ is the observed endpoint. Under conditional logistic regression,

$$\Pr(j | \mathcal{C}_t) = \frac{\exp(z_{tj}\gamma)}{\sum_{k=0}^K \exp(z_{tk}\gamma)}.$$

An SSF therefore asks which characteristics distinguished the chosen step from the alternatives reachable from the same starting point.

2.2.9 Integrated Step Selection

Empirical step-length and turning-angle distributions have already been shaped by habitat selection. Short observed steps, for example, may reflect intrinsically slow movement or prolonged residence in selected habitat. Sampling available movements directly from the observed distributions therefore risks confounding movement with selection.

An **integrated Step Selection Function** estimates the movement and selection processes jointly. Let ϕ denote the baseline movement kernel and w the habitat-selection function. The transition density is

$$p(s_{t+1} | s_t, s_{t-1}) = \frac{\phi(s_{t+1} | s_t, s_{t-1}; \theta) w(x(s_{t+1}); \beta)}{\int_{\Omega} \phi(r | s_t, s_{t-1}; \theta) w(x(r); \beta) dr}.$$

The movement kernel determines which locations can be reached and with what baseline probability. The selection function then reweights those locations according to habitat. Movement and selection together determine the next relocation.

Terms such as l , $\log l$ and $\cos \theta$ allow parameters of step-length and turning-angle distributions to be estimated alongside habitat-selection coefficients. The resulting model separates the baseline movement process from the effects of environmental conditions more explicitly than a conventional SSF (Augar et al., 2016).

2.2.10 Interpretation Through Simulation

An SSF or iSSF coefficient describes a local contrast among alternatives available from the same starting point. For two candidates a and b ,

$$\frac{\Pr(a | \mathcal{C}_t)}{\Pr(b | \mathcal{C}_t)} = \exp[\eta_{ta} - \eta_{tb}].$$

Such contrasts remain directly interpretable. The long-term consequences of the coefficients are less immediate because each selected endpoint determines the next choice set. Movement and selection therefore alter both the present choice and the locations available in the future.

Predicted utilisation distributions, residence times, crossing probabilities and home-range geometry should consequently be obtained through simulation of the complete transition process. At each step, candidate movements are drawn from the fitted movement kernel, weighted according to habitat, and sampled to produce the next location. Repetition generates trajectories whose emergent spatial patterns can be compared with held-out observations.

RSFs, SSFs and iSSFs therefore represent a progression in the treatment of availability. An RSF relates use to a broader and approximately static available distribution. An SSF conditions availability upon the animal's previous location. An iSSF jointly estimates the movement process generating those local alternatives and the habitat-selection process distinguishing among them. The appropriate framework depends upon the temporal scale of the data and the inferential question being addressed.

TOWARDS BAYESIAN INFERENCE

The preceding models commonly proceed as though recorded locations were exact, environmental covariates were known without error, individuals differed only through random sampling, and habitat-selection relationships remained constant through time. Each assumption is convenient; none is generally true. A Bayesian analysis is not the only means of relaxing them, but it provides a coherent framework in which uncertain observations, latent ecological processes, individual heterogeneity and temporal change may be represented jointly and propagated into the final inference.

Let $\mathcal{D}$ denote the observed data, $\mathcal{Z}$ the unobserved ecological quantities from which those data arose, and Θ the model parameters. Bayesian inference proceeds from the posterior distribution

$$p(\mathcal{Z}, \Theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \mathcal{Z}, \Theta)p(\mathcal{Z} \mid \Theta)p(\Theta).$$

The first term describes the observation process, the second the ecological process and the third the prior information assigned to its parameters. Rather than replacing uncertain quantities by single estimates, the posterior integrates over their plausible values. This distinction is central to three objectives of **hrHSA**: propagating uncertainty, estimating individual variation and detecting change.

3.1 Propagating Locational and Environmental Uncertainty

Telemetry records an estimate of an animal's location rather than the location itself. Let s_{ij} denote the true location of individual i at observation j , and let $\tilde{s}_{ij}$ denote the recorded GNSS fix. A simple observation model is

$$\tilde{s}_{ij} \mid s_{ij}, \Sigma_{ij} \sim \mathcal{N}_2(s_{ij}, \Sigma_{ij}),$$

where Σ_{ij} describes the magnitude and orientation of locational uncertainty. Other distributions may be substituted where errors are heavy-tailed, device-specific or otherwise non-Gaussian. The ecological model is then evaluated at the latent location s_{ij} rather than treating $\tilde{s}_{ij}$ as exact (Jonsen et al., 2005; Patterson et al., 2008; Hooten et al., 2017).

Locational inaccuracy must be distinguished from failed fixes. A recorded fix may be spatially imprecise, whereas a failed fix produces no location at all. If fix success depends upon canopy cover, topography or other habitat features, the observed relocations constitute a biased sample of the underlying path. The observation process may therefore require both a model for positional error and a model for the probability of detection or successful acquisition (Frair et al., 2004; Nielson et al., 2009).

Environmental data are likewise imperfect. Let $x^*(s, t)$ denote the environmental field relevant to the animal and let $\tilde{x}_g$ denote the value represented by raster cell g , covering area A_g . A continuous raster variable may be viewed as an imperfect spatial aggregate:

$$\tilde{x}_g = \frac{1}{|A_g|} \int_{A_g} x^*(s, t) ds + \epsilon_g,$$

where ϵ_g represents measurement, interpolation or classification error. A raster cell therefore describes an average or assigned class over an area, while the animal occupies a location within it. This difference in spatial support becomes

consequential where the environmental field varies substantially within cells or where locational uncertainty is large relative to raster resolution.

Raster uncertainty arises from several sources: sensor and classification error, temporal mismatch, interpolation, re-projection, resampling and the averaging of heterogeneous conditions within pixels. Grain size is not merely another random error term. It determines the spatial support over which habitat is represented and may therefore alter the ecological relationship being estimated. Uncertainty about grain should be addressed through biologically motivated multiscale models or sensitivity analyses rather than concealed within a single residual variance (Northrup et al., 2022).

A hierarchical model may combine these uncertainties schematically as

$$\begin{aligned}\tilde{s}_{ij} &\sim p(\tilde{s}_{ij} \mid s_{ij}, \psi_s), \\ \tilde{x} &\sim p(\tilde{x} \mid x^*, \psi_x), \\ y_{ij} &\sim p(y_{ij} \mid s_{ij}, x^*, \beta_i),\end{aligned}$$

where ψ_s and ψ_x govern locational and environmental uncertainty. The posterior distribution of β_i then reflects uncertainty not only in sampling, but also in the positions and environmental conditions upon which inference depends.

This propagation is especially important for derived quantities. Selection surfaces, utilisation distributions and simulated trajectories should not be calculated solely from posterior mean coefficients and a single environmental raster. They may instead be generated repeatedly from posterior draws of locations, environmental fields and model parameters, producing a distribution of predictions rather than one deceptively exact map.

3.2 Individual Variation as Biological Information

A population-level coefficient describes a central tendency; it does not imply that all individuals respond alike. Animals may differ in habitat selection because of sex, age, reproductive state, experience, physiology, social status, competitive ability or behavioural phenotype. Pooling these differences into a single coefficient may obscure opposing responses and understate uncertainty at the population level (Leclerc et al., 2016; Muff et al., 2020; Northrup et al., 2022).

Individual-specific selection coefficients may be represented hierarchically as

$$\beta_i = \mu_\beta + Z_i\gamma + b_i,$$

with

$$b_i \sim \mathcal{N}(0, \Sigma_\beta).$$

Here, μ_β is the population-level mean response, $Z_i\gamma$ describes systematic differences associated with measured individual attributes, and b_i represents residual variation among individuals. The covariance matrix Σ_β quantifies both the magnitude of individual variation and correlations among selection responses.

This model partially pools information across animals. Individuals with abundant and informative data retain strongly individual estimates, whereas poorly sampled individuals are drawn towards the population distribution. Partial pooling avoids the two extremes of treating every relocation as an independent population replicate and fitting entirely separate models whose imprecise estimates are subsequently treated as exact.

Individual variation also changes population-level prediction. Because the selection function is nonlinear,

$$E_i[\exp(x\beta_i)] \neq \exp[x E_i(\beta_i)].$$

The prediction obtained from the mean individual is therefore not generally equal to the mean prediction across individuals. Population-level maps should integrate over the estimated distribution of individual responses rather than merely substitute μ_β into the selection function.

Among-individual variation is not statistical debris around a population mean. It is a property of the population and may reveal individual specialisation, alternative behavioural tactics or differences in environmental sensitivity. Phenotypic

variation is also the material upon which natural selection can act. Variation in estimated habitat-selection behaviour is not, by itself, evidence of adaptive potential: an evolutionary response additionally requires that differences are sufficiently persistent, heritable and associated with fitness. Hierarchical habitat-selection models can establish and quantify behavioural variation, while pedigrees, genomic information, repeated observations and demographic data are required to distinguish its genetic, environmental and fitness-related components (Leclerc et al., 2016; Gervais et al., 2022).

3.3 Detecting Change

Conventional selection models assume that the coefficient vector β remains constant over the period of inference. Yet selection may vary with season, life-history stage, population density, resource availability, disturbance, learning or climatic conditions. In a changing environment, it is therefore insufficient to ask only which habitats are selected on average. We must also ask whether, when and among whom the relationship is changing.

A time-varying selection function may be written as

$$w_{i,t}(x) = \exp(x\beta_{i,t}).$$

Where change is expected to be gradual, the coefficients may evolve according to a stochastic process such as

$$\beta_{i,t} \sim \mathcal{N}(\beta_{i,t-1}, Q),$$

where Q controls the rate and covariance of temporal change. Smaller values of Q imply comparatively stable selection, whereas larger values permit more rapid variation. Alternative models may impose seasonal periodicity, smooth temporal trends or dependence upon measured environmental drivers.

Where an abrupt transition is expected, a change-point model may instead be used:

$$\beta_{i,t} = \begin{cases} \beta_i^{(1)}, & t < \tau_i, \\ \beta_i^{(2)}, & t \geq \tau_i, \end{cases}$$

where τ_i is the unknown time of change. Bayesian inference yields a posterior distribution for τ_i rather than forcing the transition to coincide with an analyst-defined season or calendar date.

Temporal change and individual variation may be separated through

$$\beta_{i,t} = \mu_t + b_i + u_{i,t},$$

where μ_t is the population-wide trajectory, b_i is a persistent individual deviation and $u_{i,t}$ describes within-individual change. This distinction separates population-wide redistribution from differences in individual timing and response. It also prevents changes in the composition of the sampled population from being mistaken for behavioural change within individuals.

Posterior distributions permit direct statements concerning change. For a coefficient β_j , one may calculate

$$\Pr(\beta_{j,t_2} - \beta_{j,t_1} > 0 \mid \mathcal{D}),$$

or the posterior probability that a change exceeds a biologically meaningful threshold. The result describes the magnitude, direction and uncertainty of change rather than reducing inference to a sequence of separate significance tests.

This capacity is particularly relevant under global change. Changes in climate, land use and human disturbance may alter both the distribution of available habitat and the manner in which animals respond to it. A dynamic model can distinguish a changing environmental landscape from a changing selection relationship, provided that both used and available conditions are represented through time. It may reveal gradual adjustment, abrupt displacement, increased variability or divergent responses among individuals.

Temporal change in a coefficient does not, however, identify its cause. Apparent non-stationarity may arise from season, reproduction, ageing, population density, behavioural state, changing availability or alteration of the observation process. Attribution to global change requires explicit environmental hypotheses, suitable temporal replication and, where possible, comparison across populations or landscapes. Dynamic Bayesian models expose change and quantify its uncertainty; they do not replace ecological reasoning.

3.4 Posterior Prediction and Model Evaluation

The principal practical advantage of the Bayesian formulation is that uncertainty can be carried into every derived prediction. Let Θ collect the parameters and latent states of the fitted model. The posterior predictive distribution is

$$p(\tilde{\mathcal{D}} | \mathcal{D}) = \int p(\tilde{\mathcal{D}} | \Theta) p(\Theta | \mathcal{D}) d\Theta.$$

Each posterior draw represents one plausible state of the ecological system. Repeated prediction or simulation therefore propagates uncertainty in locations, environmental covariates, individual responses and temporal dynamics into selection maps, utilisation distributions and simulated movement paths.

Bayesian inference does not remove the need for validation. Priors should be examined through prior-predictive simulation, fitted models through posterior-predictive checks, and predictive performance through the temporally blocked and leave-one-individual-out procedures described in the preceding chapter. Weak identifiability, an inappropriate availability distribution or an inadequate ecological model cannot be repaired by increasingly elaborate computation.

The purpose of the Bayesian extension is therefore not complexity for its own sake. It is to bring the inferential model into closer correspondence with the ecological system: locations and environments are observed imperfectly, populations consist of heterogeneous individuals, and selection may change through time. Representing these processes jointly permits uncertainty to remain visible, individual variation to become an object of inference and ecological change to be detected rather than averaged away.

3.5 Principal References

Auger-Méthé, M., Newman, K., Cole, D., Empacher, F., Gryba, R., King, A. A., Leos-Barajas, V., Mills Flemming, J., Nielsen, A., Petris, G. & Thomas, L. (2021). A guide to state-space modeling of ecological time series. *Ecological Monographs*, **91**, e01470.

Dejeante, R., Valeix, M. & Chamaillé-Jammes, S. (2024). Time-varying habitat selection analysis: a model and applications for studying diel, seasonal, and post-release changes. *Ecology*, **105**, e4233.

Frair, J. L., Nielsen, S. E., Merrill, E. H., Lele, S. R., Boyce, M. S., Munro, R. H. M., Stenhouse, G. B. & Beyer, H. L. (2004). Removing GPS collar bias in habitat selection studies. *Journal of Applied Ecology*, **41**, 201–212.

Gervais, L., Morellet, N., David, I., Hewison, M., Réale, D., Goulard, M., Chaval, Y., Lourtet, B., Cargnelutti, B., Merlet, J., Quéméré, E. & Pujol, B. (2022). Quantifying heritability and estimating evolutionary potential in the wild when individuals that share genes also share environments. *Journal of Animal Ecology*, **91**, 1239–1250.

Hooten, M. B., Johnson, D. S., McClintock, B. T. & Morales, J. M. (2017). *Animal Movement: Statistical Models for Telemetry Data*. Boca Raton: CRC Press.

Jonsen, I. D., Flemming, J. M. & Myers, R. A. (2005). Robust state-space modeling of animal movement data. *Ecology*, **86**, 2874–2880.

Leclerc, M., Vander Wal, E., Zedrosser, A., Swenson, J. E., Kindberg, J. & Pelletier, F. (2016). Quantifying consistent individual differences in habitat selection. *Oecologia*, **180**, 697–705.

Muff, S., Signer, J. & Fieberg, J. (2020). Accounting for individual-specific variation in habitat-selection studies: efficient estimation of mixed-effects models using Bayesian or frequentist computation. *Journal of Animal Ecology*, **89**, 80–92.

Nielson, R. M., Manly, B. F. J., McDonald, L. L., Sawyer, H. & McDonald, T. L. (2009). Estimating habitat selection when GPS fix success is less than 100%. *Ecology*, **90**, 2956–2962.

Northrup, J. M., Vander Wal, E., Bonar, M., Fieberg, J., Laforge, M. P., Leclerc, M., Prokopenko, C. M. & Gerber, B. D. (2022). Conceptual and methodological advances in habitat-selection modeling: guidelines for ecology and evolution. *Ecological Applications*, **32**, e02470.

- Patterson, T. A., Thomas, L., Wilcox, C., Ovaskainen, O. & Matthiopoulos, J. (2008). State-space models of individual animal movement. *Trends in Ecology & Evolution*, **23**, 87–94.
- Aarts, G., Fieberg, J. & Matthiopoulos, J. (2012). Comparative interpretation of count, presence–absence and point methods for species distribution models. *Methods in Ecology and Evolution*, **3**, 177–187.
- Avgar, T., Potts, J. R., Lewis, M. A. & Boyce, M. S. (2016). Integrated step selection analysis: bridging the gap between resource selection and animal movement. *Methods in Ecology and Evolution*, **7**, 619–630.
- Boyce, M. S., Vernier, P. R., Nielsen, S. E. & Schmiegelow, F. K. A. (2002). Evaluating Resource Selection Functions. *Ecological Modelling*, **157**, 281–300.
- Fieberg, J., Matthiopoulos, J., Hebblewhite, M., Boyce, M. S. & Frair, J. L. (2010). Correlation and studies of habitat selection: problem, red herring or opportunity? *Philosophical Transactions of the Royal Society B*, **365**, 2233–2244.
- Fieberg, J., Signer, J., Smith, B. & Avgar, T. (2021). A “how to” guide for interpreting parameters in habitat-selection analyses. *Journal of Animal Ecology*, **90**, 1027–1043.
- Forester, J. D., Im, H. K. & Rathouz, P. J. (2009). Accounting for animal movement in estimation of Resource Selection Functions: sampling and data analysis. *Ecology*, **90**, 3554–3565.
- Fortin, D., Beyer, H. L., Boyce, M. S., Smith, D. W., Duchesne, T. & Mao, J. S. (2005). Wolves influence elk movements: behavior shapes a trophic cascade in Yellowstone National Park. *Ecology*, **86**, 1320–1330.
- Hirzel, A. H., Le Lay, G., Helfer, V., Randin, C. & Guisan, A. (2006). Evaluating the ability of habitat-suitability models to predict species presences. *Ecological Modelling*, **199**, 142–152.
- Johnson, C. J., Nielsen, S. E., Merrill, E. H., McDonald, T. L. & Boyce, M. S. (2006). Resource Selection Functions based on use–availability data: theoretical motivation and evaluation methods. *Journal of Wildlife Management*, **70**, 347–357.
- Northrup, J. M., Hooten, M. B., Anderson, C. R. & Wittemyer, G. (2013). Practical guidance on characterizing availability in Resource Selection Functions under a use–availability design. *Ecology*, **94**, 1456–1463.
- Roberts, D. R. et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, **40**, 913–929.
- Warton, D. I. & Shepherd, L. C. (2010). Poisson point-process models solve the “pseudo-absence problem” for presence-only data in ecology. *The Annals of Applied Statistics*, **4**, 1383–1402.

IMPLEMENTATION

4.1 Design Principles

hrHSA is implemented as a scientific software package rather than as a collection of study-specific scripts. Its architecture is guided by four principles:

1. **Reproducibility** — every transformation between the original data and the final inference should be explicit and recoverable.
2. **Modularity** — data preparation, feature construction, statistical inference, validation and prediction should remain separable.
3. **Efficiency** — computational resources should be used deliberately, with particular attention to memory access, spatial chunking and repeated raster operations.
4. **Scalability** — the same analytical workflow should run on a notebook, a multicore workstation or a SLURM-managed computing cluster.

The resulting workflow separates scientific decisions from computational execution:

```
telemetry and environmental data
↓
spatial standardisation
↓
definition and sampling of availability
↓
environmental feature extraction
↓
model-matrix construction
↓
statistical inference
↓
validation and diagnostics
↓
spatial prediction or movement simulation
```

Each stage exposes a limited interface and may be inspected, tested or replaced independently.

4.2 Package Architecture

The source code follows the conventional Python `src` layout and is divided according to analytical responsibility:

```
src/hsa/
├── sampling.py          # availability domains and raster sampling
├── features.py          # reproducible construction of model matrices
├── types.py             # shared specifications and data containers
├── rsf/                 # fitting, prediction, validation and cross-validation
├── movement/           # trajectories, steps, angles and movement kernels
├── remote_sensing/     # optional Earth Observation interfaces
├── compute/             # chunking, scalable I/O and distributed execution
└── simulation/         # forward movement simulation
```

The package itself remains independent of particular species, reserves, local file systems or Earth Engine projects. Study-specific assumptions belong in configuration files, examples or downstream analyses. This distinction allows empirical workflows to remain fully documented without embedding them in the reusable statistical core.

Optional functionality is separated into dependency groups. A minimal installation provides the numerical and geospatial foundations of the package, while Earth Engine, distributed computation and HPC support may be installed only where required:

```
pip install -e .
pip install -e ".[earthengine]"
pip install -e ".[hpc,io]"
```

Optional libraries are imported lazily inside the functions that require them. Importing **hrHSA** therefore neither initialises external services nor imposes a distributed-computing environment upon smaller analyses.

4.3 Reproducible Model Construction

A fitted model is defined not only by its coefficients, but also by the transformations applied before estimation. **hrHSA** represents these transformations through a declarative `FeatureSpec`:

```
from hsa import FeatureSpec

spec = FeatureSpec(
    linear=["ndvi", "distance_to_water"],
    quadratic=["distance_to_water"],
    interactions=[("ndvi", "distance_to_water")],
    categorical=["landcover"],
    add_const=True,
)
```

Continuous predictors are standardised during fitting. Quadratic and interaction terms are then constructed on the standardised scale. Categorical levels, reference categories and final column order are stored as model metadata.

```
model, scaler, spec, meta = fit_rsf(samples, spec)
```

The model, fitted scaler, feature specification and metadata jointly constitute the fitted analysis. Passing the same objects to validation and prediction ensures that raster surfaces and held-out observations receive precisely the transformations used during model fitting.

Randomised operations accept explicit seeds. Coordinate reference systems are checked before spatial operations, and functions involving distances require projected coordinates with meaningful linear units. Invalid inputs are rejected early rather than being allowed to produce numerically plausible but ecologically incorrect results.

4.4 Geospatial Data and Scalable I/O

Telemetry locations are represented as `GeoDataFrame` objects, while environmental fields are stored as labelled `xarray` arrays. A conventional environmental stack has the dimensions

```
band × y × x
```

and retains its coordinates, projection, spatial resolution and band names throughout the workflow.

Large raster stacks may be stored as **Zarr**, preserving multidimensional structure and supporting parallel access:

```
from hsa.compute import write_raster_stack_zarr

write_raster_stack_zarr(
    env,
    "environment.zarr",
    target_chunk_mb=128,
)
```

Large sampled point tables may be stored as **Parquet**:

```
from hsa.compute import write_table_parquet

write_table_parquet(samples, "samples.parquet")
```

These intermediate products separate expensive environmental processing from statistical fitting. An analysis can therefore be repeated with a different model specification without downloading, transforming and sampling the complete environmental dataset again.

4.5 Chunking and Memory Management

High-resolution analyses commonly exceed available memory long before they exceed available computing power. **hrHSA** therefore partitions raster and point operations into manageable blocks.

For a raster containing B bands, spatial chunk dimensions n_x and n_y , and elements requiring d bytes, the approximate memory occupied by one chunk is

$$M_{\text{chunk}} = B n_x n_y d.$$

The package provides heuristics for selecting approximately square spatial chunks under a specified memory target. Point extraction is similarly divided into batches whose size depends upon the number of environmental bands and their numeric precision.

These heuristics provide conservative starting values rather than universal optima. Suitable chunk sizes depend upon the storage system, network throughput, worker memory and downstream operation. Computationally demanding analyses should therefore be benchmarked under realistic conditions.

4.6 Distributed and HPC Execution

Distributed execution is provided through **Dask**. The statistical and geospatial functions remain independent of the scheduler, allowing the same analysis to operate under different computational backends.

A local client may be created for notebook or workstation analyses:

```
from hsa.compute import make_local_dask_client

client = make_local_dask_client(
    n_workers=8,
    threads_per_worker=1,
    local_directory="/tmp/hsa-dask",
)
```

On a SLURM-managed cluster, the computational backend may be replaced without altering the scientific workflow:

```
from dask.distributed import Client
from hsa.compute import make_slurm_cluster

cluster = make_slurm_cluster(
    queue="general",
    cores=4,
    processes=4,
    memory="32GB",
    walltime="04:00:00",
    scale_jobs=10,
)

client = Client(cluster)
```

Worker-memory thresholds are configured explicitly so that intermediate data may spill to local storage before workers exhaust their available memory. Large raster stacks can be persisted across workers when they are accessed repeatedly, and temporary data can be directed to node-local scratch storage.

External services require explicit worker initialisation. Google Earth Engine, for example, maintains process-local state and must be initialised separately on each distributed worker. **hrHSA** exposes this operation directly rather than relying upon hidden state inherited from the client process.

4.7 Statistical Validation

Validation is part of the implementation rather than an optional final step. The package supports temporally blocked cross-validation and leave-one-individual-out validation, together with Boyce-index and calibration diagnostics.

Temporal blocks reduce leakage between neighbouring relocations, while leave-one-individual-out validation tests whether a model transfers to animals not represented during fitting. In the latter case, training availability is generated separately within each training animal's domain, and the held-out animal is evaluated relative to its own availability.

Validation returns fold-specific coefficients, predictions, surfaces, Boyce diagnostics, calibration tables and sample sizes. Retaining these outputs permits individual failures and unstable coefficients to be examined instead of concealing them within a single pooled score.

Movement modules provide trajectory preparation, step-length and turning-angle calculation, and estimation of empirical movement distributions by individual. These components supply the movement information required by subsequent SSF and iSSF workflows.

4.8 Bayesian and Simulation Extensions

The package architecture is intended to support hierarchical Bayesian inference and large stochastic simulation ensembles. Feature construction, environmental sampling and validation are kept independent of the fitting backend so that

likelihood-based estimation may be supplemented by PyMC- or JAX-based models without rebuilding the complete geospatial workflow.

Gradient-based algorithms such as Hamiltonian Monte Carlo and the No-U-Turn Sampler are particularly suitable for the high-dimensional and correlated parameter spaces produced by hierarchical habitat-selection models (Hoffman & Gelman, 2014; Betancourt, 2017). Independent chains can be distributed across processes or nodes, while automatic differentiation and just-in-time compilation may accelerate repeated likelihood evaluation.

Forward simulations provide a second level of parallelism. Independent trajectories or posterior simulations may be distributed among workers, while the candidate steps within each trajectory can be evaluated through vectorised array operations.

The present package provides the geospatial, movement and distributed-computing foundations for these extensions. Fully integrated Bayesian fitting and habitat-biased forward simulation remain developing components and are kept distinct from the stable RSF implementation.

4.9 Documentation and Quality Assurance

The documentation is built with **Sphinx** and **MyST Markdown**, providing mathematical notation, automatically generated API references and versioned narrative documentation. Package configuration is maintained in `pyproject.toml`, with explicit core, optional and development dependencies.

The current test suite verifies the public import surface and provides a foundation for continued quality assurance. Release hardening should additionally include:

- unit tests for spatial and statistical functions;
- regression tests for stable numerical outputs;
- synthetic parameter-recovery experiments;
- end-to-end workflow tests;
- and benchmarks of runtime, memory use and scaling efficiency.

Scientific verification must accompany software verification. A function may execute correctly while implementing an inappropriate availability domain, temporal scale or ecological model.

4.10 Reproducibility

A reproducible **hrHSA** analysis should preserve:

- package and dependency versions;
- coordinate reference systems and raster grains;
- the definition of availability;
- random seeds and sampling ratios;
- the feature specification and preprocessing metadata;
- fitted models and diagnostic outputs;
- cross-validation partitions;
- and the environmental products, or immutable references, from which predictors were derived.

Reproducibility is therefore treated as part of the analytical design rather than as an administrative addition made after the analysis is complete.

4.11 Summary

hrHSA combines an accessible Python interface with a modular architecture for high-resolution telemetry and environmental data. Declarative model specifications, labelled geospatial arrays, memory-aware batching, chunked storage, structured validation and interchangeable local or SLURM-backed execution allow the same scientific workflow to operate across different computational scales.

The purpose of this architecture is not parallelism for its own sake. It is to ensure that increasing spatial resolution, study extent and model complexity does not require a corresponding loss of transparency, reproducibility or ecological rigour.

4.12 Principal References

Betancourt, M. (2017). A conceptual introduction to Hamiltonian Monte Carlo. *arXiv*, 1701.02434.

Hoffman, M. D. & Gelman, A. (2014). The No-U-Turn Sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, **15**, 1593–1623.

Hoyer, S. & Hamman, J. J. (2017). xarray: N-D labelled arrays and datasets in Python. *Journal of Open Research Software*, **5**, 10.

Rocklin, M. (2015). Dask: parallel computation with blocked algorithms and task scheduling. *Proceedings of the 14th Python in Science Conference*, 130–136.

Sandve, G. K., Nekrutenko, A., Taylor, J. & Hovig, E. (2013). Ten simple rules for reproducible computational research. *PLoS Computational Biology*, **9**, e1003285.

Wilkinson, M. D. et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, **3**, 160018.

Wilson, G. et al. (2017). Good enough practices in scientific computing. *PLoS Computational Biology*, **13**, e1005510.

CASE STUDIES

5.1 Static RSF (Pangolin)

5.2 Seasonally-Variant RSF (Lion)

5.3 Dynamic SSF (Vultures)

API REFERENCE